# Rob Romijnders

PhD in Machine Learning

## About me

PhD student in ML.

Specializing in large-scale and robust machine learning.

## Community engagement

Recorded talks at PyData:

- Machine Translation (2017)
  youtube.com/watch?v=HVdPWoZ_swY
- Bayesian ML (2018)
  youtube.com/watch?v=Z7VN7oRA6TY
- AI robustness (2024)
  youtube.com/watch?v=2-EjNJnV4Ec

## Languages

- Dutch/English: native
- French: intermediate
- German: intermediate

## Experience

Sep-2021 / present

### PhD in Machine Learning

University of Amsterdam · Amsterdam, NL

- My main topic is federated machine learning and statistical inference;
- Advised by Max Welling, Christos Louizos, and Yuki M. Asano;
- Thesis supervisor to three cum-laude graduates;
- Sponsored by Qualcomm AI, part of EU-wide ELLIS program.

**Internships during the PhD:**

- **G-Research** (London, 2024), on financial time series modeling;
- **Brave Software** (London, 2024), on robust Large Language Models (LLM);
- **Apple** (Cambridge, 2025), on scalable and robust histogram aggregation.

Oct-2019 / Jul-2021

### AI Resident Researcher

Google · Zurich, CH

I dealt with 300+ terabytes of video data on distributed file systems, had more than 180 accepted pull requests, and created and shared 60+ internal documents. This resulted in five publications in top-tier publication venues.

Nov-2016 / Jul-2019

### ML engineer

Frosha (AI startup) · Amsterdam, NL

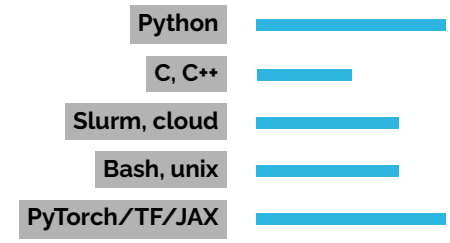
I was the main machine learning engineer in this startup, training NLP machine learning algorithms for classification and parsing of unstructured text.

## Extra curricular

**Academic Reviewing**
Reviewer at CVPR/ICLR/ICML/NeurIPS
Outstanding reviewer award ICCV 2021

**Summer schools**
GPSS summer school, UK 2018
DLRL summer school, Canada 2023
FoMo summer school, NL 2024

## Programming




## Education

2015–2018

### MSc Electrical Engineering

Eindhoven University of Technology · Eindhoven, NL

Graduated *cum laude*, top 10% of my class. Courses in signal processing, stochastic processes, dynamical systems and control theory.
*Six month exchange studies at the National University of Singapore (NUS).*

2011–2015

### Bachelor of Science

Twente University · Enschede, NL

Graduated *cum laude*, top 10% of year; board member at AIESEC Twente.

Rob Romijnders romijndersrob@gmail.com @ *robromijnders.github.io*

# Academic publications

## Robust AI and Decentralized inference

2025 **Multi-reference alignment for MIMO wireless communications**
**RR**, Cesa, Louizos, Pratik, Behboodi · NeurIPS 2025 workshop
*This project resulted in a patent filing by Qualcomm Inc.*

2025 **NoEsis: A Modular LLM with Differentially Private Knowledge Transfer**
**RR**, Laskaridis, Shahin-Shamsabadi, Haddadi · ICLR 2025 workshop
*Internship project at Brave; Code available at github.com/RobRomijnders/noesis;*
*This project involved training multiple Large Language Models for both text and code.*

2025 **Convex Approximation of Two-Layer ReLU Networks for Hidden State DP**
**RR**, Koskela · NeurIPS 2025
*This international collaboration grew out of a serendipitous meeting at a conference*

2024 **DNA: Differentially private Neural Augmentation for contact tracing**
**RR**, Louizos, Asano, Welling · ICLR 2024 workshop
*Code available at github.com/RobRomijnders/dna; awarded spotlight talk at the workshop.*

2024 **Protect Your Score: Contact Tracing with Differential Privacy Guarantees**
**RR**, Louizos, Asano, Welling · AAAI 2024
*Code available at github.com/RobRomijnders/dpfn_aaai; awarded oral, for top 10% of papers.*

2023 **No time to waste: practical statistical contact tracing with few low-bit messages**
**RR**, Asano, Louizos, Welling · AISTATS 2023
*Code available at github.com/QUVA-Lab/nttw; required decade-old math from the ML literature.*

## Large-scale AI and video data

2022 **Beyond transfer learning: Co-finetuning for action localisation**
Arnab, Xiong, Gritsenko, **RR**, Djolonga, Dehghani, Sun, Lucic, Schmid · arXiv preprint
*I wrote an important dataloader, which involved C programming and multi-node preprocessing. I trained LLMs with both images and videos for this project.*

2021 **Representation learning from videos in-the-wild: An object-centric approach**
**RR**, Mahendran, Tschannen, Djolonga, Ritter, Houlsby, Lucic · IEEE WACV 2021
*My starting project in the AI residency; this involved handling millions of videos on remote file systems.*

## Robustness, calibration, and generalization

2021 **Impact of aliasing on generalization in deep convolutional networks**
Vasconcelos, Larochelle, Dumoulin, **RR**, Le Roux, Goroshin · ICCV 2021

2021 **Revisiting the Calibration of Modern Neural Networks**
Minderer, Djolonga, **RR**, Hubis, Zhai, Houlsby, Tran, Lucic · NeurIPS 2021
*A cross-team collaboration; this involved evaluating 100+ models, including LLMs, on 50+ datasets.*

2021 **On Robustness and Transferability of Convolutional Neural Networks**
Djolonga, Yung, Tschannen, **RR**, Beyer, Kolesnikov, Puigcerver, Minderer, D'Amour, Moldovan, Gelly, Houlsby, Zhai, Lucic · CVPR 2021
*I wrote code for several evaluation setups for multiple models on the eight eval metrics.*

2019 **Data Selection for training CNNs with cross-dataset weak supervision**
Meletis, **RR**, Dubbelman · IEEE ITSC 2019

2019 **Domain Agnostic Normalization for Unsupervised Adversarial Domain Adaptation**
**RR**, Meletis, Dubbelman · IEEE WACV 2019
*This was my MSc thesis project with which I graduated cum laude.*

## AI applications

2016 **Applying Deep Learning to Basketball Trajectories**
Shah, **RR** · Sports Analytics Workshop, KDD 2016
*An international collaboration; my first academic publication right after my BSc thesis.*